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# Multi-omics–based machine learning model predicts response and guides treatment in Crohn disease: a case study in nutritional therapy

**Asaf Azulay, MSc[1], Leora Gotesdyner, MD[1], Yonat Aharoni-Frutkoff, MSc[1], Gili Focht, MSc[1], Yael Talmor, BSc[2], Elhanan Borenstein, PhD[2,3,4], Luba Plotkin, MSc[1], Esther Orlanski-Meyer, MD[1], Raffi Lev-Tzion, MD[1], Oren Ledder, MD[1], Dotan Yogev, MD[1], Amit Assa, MD[1], Efrat Broide, MD[5], Anat Yerushalmy-Feler, MD[6,7], Jarosław Kierkuś, MD[8], Tobias Schwerd, MD[9], Eytan Wine, MD, PhD[10], Dan Turner, MD, PhD[*,1]**

[1]Juliet Keidan Institute of Pediatric Gastroenterology Hepatology and Nutrition, The Eisenberg R&D Authority, Shaare Zedek Medical Center, The Hebrew University of Jerusalem, Jerusalem Israel
[2]School of Computer Science, Tel Aviv University, Tel Aviv, Israel
[3]Faculty of Medicine, Tel Aviv University, Tel Aviv, Israel
[4]Santa Fe Institute, Santa Fe, NM, United States
[5]Pediatric Gastroenterology Unit, Assaf Harofeh Medical Center, Zerifin, Israel
[6]Pediatric Gastroenterology Institute, "Dana-Dwek" Children's Hospital, Tel Aviv Sourasky Medical Center, Tel Aviv, Israel,
[7]Faculty of Medical and Health Sciences, Tel Aviv University, Tel Aviv, Israel
[8]Pediatric Gastroenterology Unit, Children's Memorial Health Institute, Warsaw, Poland
[9]Department of Pediatrics, Dr. von Hauner Children's Hospital, University Hospital, LMU Munich, Germany
[10]Division of Gastroenterology, Hepatology, and Nutrition, Hospital for Sick Children and University of Toronto, Toronto, Ontario, Canada
[*]Corresponding author: Dan Turner, MD, PhD, The Juliet Keidan Institute of Pediatric Gastroenterology and Nutrition, Shaare Zedek Medical Center, The Hebrew University of Jerusalem, P.O.B. 3+235, Jerusalem, 91031, Israel (turnerd@szmc.org.il).

## Abstract

***Background:*** **Biomarkers are needed to predict treatment response and guide therapeutic decisions in Crohn disease (CD). We aimed to develop and validate a multi-omics machine learning (ML) model to predict response to nutritional therapy in pediatric CD.**

***Methods:*** **Treatment-naive children with newly diagnosed CD who were initiating exclusive enteral nutrition (EEN) were prospectively enrolled in this study. Metabolomics and lipidomics were measured in the serum and stool, as well as the fecal microbiome. Following feature selection via minimum redundancy maximum relevance, random-forest models were constructed for single- and multi-omics and performances were evaluated. The models were externally validated in an independent prospective cohort of treatment-naive children and young adults with CD treated with EEN.**

***Results:*** **The discovery cohort consisted of 50 children (mean ± SD age 14.3 ± 2.7 years), of whom 34 (68%) responded to EEN. Combining complementary signals from host metabolism, gut microbiota, and lipid profiles from serum and stool in a multi-omics ML model yielded a model for predicting treatment response (training accuracy 94%; 95% CI, 82%-100%). Key predictive features included serum metabolites (2-hydroxyglutaric acid, Cer[d18:0/22:0], and HexCer[d18:1/d26:1]), fecal metabolites (3-methyladipic acid, DG[16:0 20:0], PC aa C42:2), and microbial taxa (family *Bifidobacteriaceae* and genus CAG-56). The validation cohort consisted of 21 patients of whom 12 (57%) responded to EEN. The multi-omics model performance achieved an area under the receiver operating characteristic curve (AUROC) of 0.81 (95% CI, 0.6-1.0). Clinical and endoscopic features did not improve the predictive ability of the model.**

***Conclusion:*** **As a proof-of-concept, we showed that integrated multi-omics ML models can predict EEN response in pediatric CD patients, supporting their potential use in precision nutrition and personalized care strategies.**

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**Key words:** Crohn disease; Exclusive enteral nutrition; Children; Prediction; Multiomics; Microbiome; Nutritional therapy; Metabolomics

## Lay Summary

**This study developed and externally validated a multi-omics machine-learning model integrating serum and fecal metabolomics, lipidomics, and microbiome data to predict pediatric Crohn disease response to exclusive enteral nutrition, achieving high accuracy. This study advances personalized medicine in pediatric IBD.**

### Key Messages

*What is already known?*

Exclusive enteral nutrition (EEN) is an effective induction therapy in pediatric CD, but response rates vary and reliable biomarkers to predict outcomes are lacking. Prior studies explored fecal metabolomics and/or microbiome profiles to predict response; these were limited by lack of external validation.

*What is new here?*

This study presents what is to our knowledge the first externally validated machine-learning model integrating serum and fecal metabolomics, lipidomics, and microbiome data to predict response in pediatric CD. The model identifies reproducible, biologically meaningful predictors with promising accuracy, requiring further exploration.

*How can this study help patient care?*

Our findings support use of multi-omics–based prediction models to guide personalized treatment, in particular, for nutritional therapy to assist in early therapeutic decision-making.

# Introduction

Increasing attention is being directed toward molecular profiling for capturing host and microbial signatures associated with individual therapeutic response. Multi-omics technologies, such as metabolomics, lipidomics, and microbiome, provide a comprehensive view of the immuno-metabolic and microbial landscape in inflammatory bowel diseases (IBD). These methods hold promise for developing molecular classifiers that could guide personalized therapy.[1–3] However, these datasets are high-dimensional and often hard to interpret due to high technical variability, missing data, and interpatient heterogeneity.

Meanwhile, machine learning (ML) has gained momentum in IBD research as a robust strategy for extracting patterns from complex biological datasets, enabling the integration of diverse data types, and ML has been applied for developing prediction models.[4,5] Despite growing interest, the use of multi-omic ML models in IBD has been limited by practical challenges of small cohort sizes, heterogeneous data quality, and the lack of independent validation, particularly in pediatrics.[6]

Exclusive enteral nutrition (EEN) is a well-established first-line induction therapy for pediatric Crohn disease (CD), offering high remission rates and improved nutritional status.[7,8] However, 30%-40% of children fail to respond to EEN, but current clinical, endoscopic, or laboratory parameters cannot reliably predict which patients will benefit.[9–11] This presents a gap in early treatment decision-making, during which timely redirection to alternative therapies may improve long-term outcomes and prevent the emotional consequences of challenging nutritional treatment in futile cases.

A recent study by Nichols et al.[12] demonstrated the potential of fecal metabolomics and microbiome to predict EEN response in pediatric CD, identifying distinct profiles in responders at baseline. Alongside this study a previous study showed the potential of microbial species and clinical data in predicting EEN response.[13] While the findings underscore the clinical potential of omics-based approaches, these studies included only fecal omics layers and lacked external validation.

In this exploratory study, we aimed to develop and validate a multi-omics–based ML model integrating serum and fecal metabolomics, lipidomics, gut microbiome, and clinical data to predict response to EEN in CD. We demonstrated the added value of integrative approach over single-modality data for supporting timely personalized treatment decisions.

# Methods

## Discovery cohort

Children newly diagnosed with CD (age 0-18 years) were enrolled from 2013 to 2022 into a prospective cohort at Shaare Zedek Medical Center (SZMC) prior to initiating treatment with the goal of identifying predictors of response to treatments and disease course. All patients were enrolled under a uniform protocol, with identical procedures for sample collection, processing, and analysis, ensuring methodological consistency. For this study we included children initiating EEN as the sole induction treatment. The clinical management of EEN at SZMC remained standardized according to European Society for Paediatric Gastroenterology Hepatology and Nutrition (ESPGHAN)/European Crohn's and Colitis Organisation (ECCO) guidelines throughout the enrollment period. We excluded patients receiving partial enteral nutrition, and those not completing at least 2 weeks of treatment, a minimal time period required to assess biologic response rather than adherence. We also excluded children initiating other medical treatments in addition to the nutritional intervention except for thiopurine initiation because it takes at least 2 months for these drugs to become effective; children who had been treated with thiopurines for more than 2 months were excluded.

Serum and stool samples were collected in parallel to recording explicit clinical, laboratory, and demographic data on a REDCap electronic case report form. Disease activity was quantified by standard indices of colonoscopic degree of inflammation (using the Simple Endoscopic Score in CD [SES-CD][14]), disease classifications (using IBD-classes and Paris criteria[15]), and validated multiitem indices (ie, weighted Pediatric CD Activity index [wPCDAI][16] and the Mucosal Inflammation Non-Invasive [MINI] index[17]). Children were prospectively followed forpostintervention outcomes.

## Validation cohort

As an independent external validation cohort, we utilized the EEN arm of a multicenter randomized controlled trial that compared EEN with the Tasty&Healthy exclusive whole-food diet (NCT#04239248,[18] patient enrollment between 2020 to 2023). Briefly, biologic-naive children and young adults (aged 6-25 years) with mild to moderately active CD were randomized to the EEN or Tasty&Healthy intervention for 8 weeks. Participants were seen at baseline, week 4, and week 8 for assessment and for collection of clinical records, serum, and stool. Samples were collected and analyzed for the different omics using the same operating procedures, technologies, and laboratories as the discovery cohort. As with the discovery cohort, we included those treated with EEN for at least 2 weeks.

## Outcomes

The primary outcome was treatment response, defined as either clinical remission (ie, wPCDAI score $< 12.5$) in those completing at least 6 weeks of the 8-week intervention, or clinical response (ie, reduction of $> 17.5$ points in wPCDAI from week 0) in those discontinuing EEN or initiating other treatments after 2-6 weeks; those not completing at least 2 weeks of intervention were excluded as this duration was the minimal for assessing biologic response to the intervention.[16] For 3 patients wPCDAI could not be calculated due to missing core components and their response was determined based on the documented clinical assessment of the treating physician, supported by available objective findings, as reviewed by 2 independent clinicians (D.T. and Y.A.F.) who were blinded to the multi-omics results throughout this process.

## Sample handling and omics data generation

Stool samples for metabolomics and lipidomics were collected at disease onset prior to treatment, divided into 5-mL tubes, and stored at −80 °C until further analyses. Stool for microbiome analysis was fixated immediately in tubes containing 95% ethanol, centrifuged to pellet stool, of which 100-200 mg was sub-aliquoted into 2-mL cryovials and stored at −80 °C. Blood was collected in serum separator tube tubes and centrifuged, and serum was aliquoted and stored at −80 °C for metabolomics. Serum and stool metabolomics and lipidomics data were generated in both cohorts in the Metabolomics Innovation Centre (TMIC), University of Alberta (Edmonton, Canada). Details of the analyses can be found in the Supplementary Methods. Briefly, for metabolomics, a targeted quantitative approach was used for direct injection (for metabolites) or ultra-high-performance liquid chromatography (UHPLC; for bile acids) with mass spectrometry (MS) or with liquid chromatography–MS/MS. Metabolites and lipids which were below the limit of detection (LOD) values in $> 75\%$ of samples were excluded to eliminate unreliable signals. The remaining values under the LOD were imputed as half of the LOD value; no other imputation was performed in this study. The data were split for analyses into metabolites and bile acids combined (herein metabolomics) and lipids (herein lipidomics). To equalize signals of metabolite concentrations, data were log-transformed and scaled using auto *z* score normalization. For the microbiome, DNA was extracted from stool and the 16S rRNA gene was sequenced. Libraries were cleaned and quantified and equimolar amounts were pooled and sequenced to generate paired-end reads. Raw sequences were processed using the DADA2 standard workflow, followed by taxonomic assignment, rarefication, filtration, and normalization, resulting in normalized taxonomy tables (see Supplementary Methods).

A multi-omics model was then sought while integrating the following: (1) demographics (ie, age, body mass index [BMI] and sex), baseline disease activity markers (eg, wPCDAI and SES-CD), disease location (L1 or L2/L3; L4b was grouped with L4a given the small number of children with the L4b phenotype) and behavior (B1 or B2; there was 1 participant with the B2B3 phenotype, which was grouped with B2); (2) serum metabolomics; (3) serum lipidomics; (4) fecal metabolomics; (5) fecal lipidomics; and (6) fecal microbiome. The multi-omics data were integrated using an early integration approach, which involved a concatenation of features across all previously generated and processed omics data sets.[19] This method combined clinical and omics features into a unified dataset. By merging the different data types at an early stage, we were able to capture potential interactions and dependencies while providing a holistic view of the underlying biological system.

## Analytic approach

Univariate analyses were performed with the *t*-test or Wilcoxon rank sum test, as appropriate, while correcting for multiple comparisons using the Benjamini Hochberg false discovery rate (FDR; implemented in the statsmodels package in python). For the microbiome analyses, α-diversity, which quantifies the intra-sample diversity, was estimated using chao1 richness and Shannon diversity index, while β-diversity, which measures dissimilarities between participants, was calculated using the Bray-Curtis dissimilarity index. Groups were compared using the Kruskal Wallis test, principal coordinate analysis (PCoA), and Permutational Multivariate Analysis of Variance (PERMANOVA), except for differential abundance that was assessed using Microbiome Multivariable Associations with Linear Models (MaAsLin2).[20]

Machine learning enables the recognition of complex patterns, providing a dynamic approach to outcome prediction in heterogeneous datasets. Using the discovery cohort, random forest (RF) classifiers comprising 500 decision trees were fitted with leave-one-out cross-validation (LOOCV) and training performance was evaluated, first on each single-omics dataset individually (ie, clinical variables, serum metabolomics and lipidomics, fecal metabolomics and lipidomics, and microbiome), and subsequently on the integrated multi-omics subset. Given the limited cohort size and the exploratory nature of this study, LOOCV was chosen to maximize training data usage in each iteration and to ensure that every sample served once as an independent test case. As an initial preprocessing step, log transformation was applied to each omics dataset separately. Within each train-test iteration, data were standardized using *z* score normalization, and feature selection was performed using the minimum redundancy maximum relevance (mRMR) algorithm.[21] This algorithm identifies the most informative variables by maximizing their relevance to the outcome while minimizing redundancy among selected features, achieved by penalizing correlated variables based on mutual information. In each split iteration, the top 5 contributing features were retained. The most frequently retained features across LOOCV folds were then used to fit a final RF model on the discovery cohort for each single-omics layer to assess model training performance. For multi-omics integration and model development, the retained features from all single-omics analyses were combined. As an optimization step to further minimize redundancy and enhance interpretability, systematic recursive feature elimination was applied using only the multi-omics subset in the discovery cohort. The final single-omics and multi-omics models were then evaluated using identical metrics and methodology on an independent external validation cohort to assess generalizability. All model parameters and selected features were fixed prior to testing on the validation cohort, which remained completely unseen during model development and optimization. Feature importance was quantified using normalized Gini importance, defined as the total reduction in impurity contributed by each variable across all trees. To facilitate comparison, raw importance values were normalized to sum to 1.[22] Model performance was assessed using the area under the receiver operating characteristic curve (AUROC), accuracy, sensitivity, specificity, and positive and negative predictive values (PPV, NPV). The optimal classification threshold was determined with the Youden's J statistic, which maximizes the sum of sensitivity and specificity. 95% CIs were estimated via bootstrapping. Principal component analysis (PCA) was used to visualize sample-level variation in a 2-dimensional space defined by the first 2 principal components, with ellipses drawn around group centroids (responders and nonresponders) to illustrate within-group dispersion (see Supplementary Methods for details). All analyses were performed in Python 3.8 using the packages *scipy*, *scikit-learn*, *statsmodels*, *pingouin*, *mrmr*, *matplotlib*, and *seaborn*, except for microbiome data, which were analyzed in R 4.3.3 using the packages *vegan*, *phyloseq*, and *MaAsLin2*. The study was approved by the institutional ethics committee, and written informed consent was obtained from all participants.

# Results

## Overview of cohorts

A total of 50 children met the eligibility criteria of the discovery cohort, of whom 34 (68%) were responders (Table 1). Serum metabolomics and lipidomics were available for 34 children (68%), fecal metabolomics and lipidomics for 19 (38%), and fecal microbiome for 31 (62%) children; 17 children (34%) had all components (Figure S1). The reasons for missing data included refusal to provide stool or blood samples, collection after initiation of EEN, or insufficient amount. The validation cohort included 21 children and young adults, all of whom had all omics components; of those, 12 (57%) were responders (Table 1).

## Clinical data

Clinical, biochemical, endoscopic and demographic variables were recorded for all patients in both cohorts at baseline, but none predicted response to EEN in either training performance in the discovery cohort or in the validation cohort (Table S1 and Table 2).

## Single-omics in the discovery cohort

### Serum and fecal metabolomics and lipidomics- univariate analysis

A total of 138 serum metabolites, 433 serum lipids, 172 fecal metabolites, and 185 fecal lipids were retained after data processing and feature reduction. Univariate analysis of traditional statistics showed differential abundance of several metabolites and lipids between responders and nonresponders (Figure S2). Eight *serum metabolites* were more abundant in nonresponders (hippuric acid, malonic acid, caproic acid, 3-carboxy-4-methyl-5-propyl-2-furanpropionic acid [CMPF], 3-methyladipic acid, alpha-aminobutyric acid, butyric acid + isobutyric acid, and N-acetylproline), while 2 were less abundant (putrescine and 2-hydroxyglutaric acid; Table S2, Figure S1A). Several *fecal metabolites* were also differentially abundant between responders and nonresponders. One fecal metabolite was more abundant in nonresponders (alpha-aminobutyric acid) and 3 were less abundant (gamma-aminobutyric acid, 3-methyladipic acid and orotic acid; Table S3, Figure S1B). Eight *serum lipids* were more abundant in nonresponders, including ceramides, cholesterol esters, diglycerides, and triglycerides, while 3 lipids, including dodecanoylcarnitine, butenylcarnitine, and a diglyceride, were less abundant (Table S4, Figure S1C). Two *fecal lipids* were differentially abundant, a diglyceride and a phosphatidylcholine, which were more abundant in nonresponders (Table S5, Figure S1D).

### Serum and fecal metabolomics and lipidomics-ML

None of the metabolites or lipids in the aforementioned univariate analyses reached statistical significance after FDR correction for multiple comparisons (Table S2-5), likely since the sample size was insufficient for comparing the groups by traditionally-corrected statistical comparisons. Therefore, we generated a ML RF model for each dataset separately utilizing the top 5 retained features by the mRMR algorithm. Reassuringly, the feature selection algorithm mostly retained features that were significant or nearly significant in the traditional univariate analyses. The *serum lipidomics* model retained 3 ceramides (Cer(d18:0/22:0), Cer(d18:0/24:0) and HexCer(d18:1/d26:1)) and 2 acylcarnitines (dodecanedioylcarnitine [C12DC] and valerylcarnitine [C5]). Dodecanedioylcarnitine was more abundant in responders, while the others were enriched in nonresponders (Figure 1A). In the *fecal lipidomics* model, predictive features included PC aa C42:2, DG(16:0 20:0), and PC aa C40:5, which were higher in nonresponders, and PC aa C42:4 and Propionylcarnitine (C3), which were elevated in responders (Figure 1B). These features yielded RF models with moderate training performance (serum: AUROC 0.77, 95% CI, 0.58-0.94; accuracy, 74% 95% CI, 0.59-0.88 Fecal: AUROC 0.67 95% CI, 0.27-0.98, accuracy 74% 95% CI, 0.53-0.89; Table S1).

**Table 1** Baseline characteristics of the included cohorts

| | Discovery cohort | | | Validation cohort | | |
|---|---|---|---|---|---|---|
| | Nonresponders (*n* = 16) | Responders (*n* = 34) | *P* value | Nonresponders (*n* = 9) | Responders (*n* = 12) | *P* value |
| **Sex, males, *n* (%)** | 9 (56%) | 21 (62%) | .95 | 3 (33%) | 6 (50%) | .75 |
| **Age, y, mean ± SD** | 14.1 ± 2.8 | 14.4 ± 2.7 | .67 | 13.7 ± 3.1 | 14.3 ± 3.0 | .65 |
| **wPCDAIm, median (IQR)** | 56.3 (45.6-65.6) | 55 (40.6-72.5) | .84 | 42.5 (32.5-50) | 37.5 (31-46.2) | .52 |
| **Severe, *n* (%)** | 8 (50%) | 16 (47%) | | 2 (22%) | 1 (8%) | |
| **Moderate, *n* (%)** | 6 (38%) | 10 (29%) | | 4 (45%) | 5 (42%) | |
| **Mild, *n* (%)** | 2 (12%) | 8 (24%) | | 3 (33%) | 6 (50%) | |
| **CRP, median (IQR), mg/dL** | 5.4 (2.1-17.8) | 5.6 (2.8-34.2) | .58 | .8 (0.4-2.8) | 2.4 (1.1-3.5) | .3 |
| **SES-CD, median (IQR)** | 16.0 (7.8-21.0) | 15.0 (8.2–18.0) | .72 | – | – | – |
| **Disease behavior**[a] | | | .26 | | | 1.0 |
| **B1, *n* (%)** | 12 (75%) | 30 (88%) | | 9 (100%) | 11 (92%) | |
| **B2/B3, *n* (%)** | 4 (25%) | 4 (12%) | | 0 (0%) | 1 (8%) | |
| **Disease location**[b] | | | .90 | | | .78 |
| **L1** | 4 (25%) | 12 (35%) | | 6 (67%) | 5 (42%) | |
| **L2** | 0 (0%) | 0 (0%) | | 0 (0%) | 1 (8%) | |
| **L3** | 12 (75%) | 22 (65%) | | 3 (33%) | 6 (50%) | |
| **L4** | 8 (50%) | 18 (53%) | | 3 (33%) | 5 (42%) | |

Abbreviations: BMI, body mass index; CRP, C-reactive protein; EEN, exclusive enteral nutrition; SES-CD, Simple Endoscopic Score in Crohn's Disease; wPCDAI, weighted pediatric Crohn's disease activity index.

[a] According to the Montreal classification.

[b] According to the Paris classification.

**Table 2** Evaluation metrics in external validation cohort for random forest models of the individual omics and the multi-omics.[a]

| Metric/ dataset[b] | Multi-omics | Clinical | Serum metabolites | Serum lipids | Fecal metabolites | Fecal lipids | Microbiome |
|---|---|---|---|---|---|---|---|
| **AUROC** | 0.81 (0.55-1.0) | 0.48 (0.22-0.76) | 0.24 (0.0-0.51) | 0.74 (0.47-0.95) | 0.5 (0.21-0.79) | 0.71 (0.46-0.93) | 0.56 (0.29-0.89) |
| **Accuracy** | 0.81 (0.62-0.95) | 0.48 (0.29-0.67) | 0.57 (0.38-0.76) | 0.76 (0.57-0.95) | 0.62 (0.38-0.81) | 0.67 (0.43-0.86) | 0.67 (0.48-0.86) |
| **Sensitivity** | 0.56 (0.25-0.88) | 0.89 (0.62-1.0) | 0.0 (0.0-0.0) | 0.44 (0.11-0.78) | 0.56 (0.17-0.88) | 0.44 (0.11-0.75) | 0.33 (0.0-0.67) |
| **Specificity** | 1.0 (1.0-1.0) | 0.17 (0.0-0.42) | 1.0 (1.0-1.0) | 1.0 (1.0-1.0) | 0.67 (0.38-0.92) | 0.83 (0.6-1.0) | 0.92 (0.73-1.0) |
| **PPV** | 1.0 (1.0-1.0) | 0.44 (0.22-0.67) | 0.0 (0.0-0.0) | 1.0 (1.0-1.0) | 0.56 (0.2-0.88) | 0.67 (0.25-1.0) | 0.75 (0.0-1.0) |
| **NPV** | 0.75 (0.53-0.94) | 0.67 (0.0-1.0) | 0.57 (0.38-0.76) | 0.71 (0.5-0.93) | 0.67 (0.38-0.92) | 0.67 (0.42-0.88) | 0.65 (0.41-0.87) |

Abbreviations: AUROC, area under receiver operating characteristic; NPV, negative predictive value; PPV, positive predictive value.

[a] Each column represents evaluation metrics for a machine learning random forest model created using selected components by minimum redundancy maximum relevance algorithm in validation cohort, with mean (95% CI) calculated using bootstrapping.

In *serum metabolomics,* all retained features apart from 2-hydroxyglutaric acid were higher in nonresponders while in *fecal metabolomics* most features retained by the mRMR algorithm were more abundant in responders. These included serum 2-hydroxyglutaric acid, hippuric acid and alpha-aminobutyric acid, and fecal gamma-aminobutyric acid and 3-methyladipic acid (Figure 1C-D). Both metabolomics models showed good training performance (serum: AUROC, 0.84, 95% CI, 0.69-0.96; accuracy 74%, 95% CI, 0.56-0.88 Fecal: AUROC 0.90, 95% CI, 0.72-1.00; accuracy 84%, 95% CI, 0.68-1.0).

## Microbiome- univariate analysis

A total of 201 microbial taxonomies were retained in the microbial dataset after processing. Higher Shannon index diversity (α diversity) was observed in nonresponders ($P = .03$, Figure 2A). A similar trend was observed in chao1 richness index ($P = .24$, Figure 2B) and no difference was seen in β diversity (PERMANOVA test, $F_{1,30} = 0.66$, $P = .84$; Figure 2C).

Using standard linear models, differential abundance analysis between responders and nonresponders included family *Bifidobacteriaceae* ($P = .04$), which were higher in responders and genera *Colidextribacter, Lachnospiraceae ND3007 group* and CAG-56 ($P = .01$, .02 and .05) in addition to family *Prevotellaceae* ($P = .03$) that were higher in nonresponders (Table S6, Figure 2D).

## Microbiome- ML

Reassuringly, most of the aforementioned differentially abundant taxa were also retained in the feature selection process for ML RF model training (Figure 2E). The RF model moderately

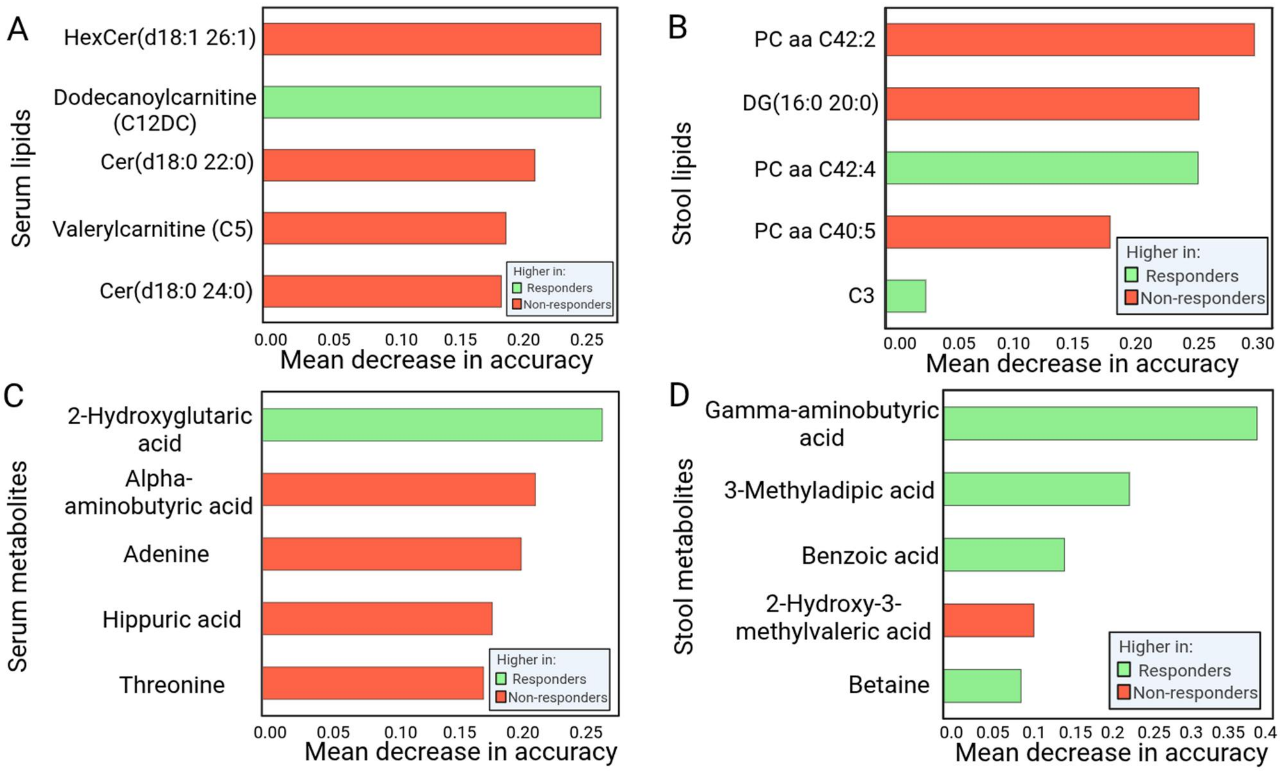


**Figure 1** Serum and fecal metabolomics and lipidomics key features from single-omics models for predicting response to EEN in CD. (A-D) Feature importance calculated via machine-learning random forest model of the features retained by the mRMR algorithm for (A) serum, (B) fecal lipidomics, (C) serum, and (D) fecal metabolomics. In the bar plots, green means higher median values in responders, and red higher median values in nonresponders. CD, Crohn disease; EEN, exclusive enteral nutrition; mRMR, minimum redundancy maximum relevance.

differentiaed responders from nonresponders (AUROC, 0.76; 95% CI, 0.57-0.92; accuracy 74%; 95% CI, 58%-87%; Table S1). Using the same retained features, a similar trend was observed in the PCA plot, in which a trend toward separation between EEN responders and nonresponders was evident (Figure 2F).

## Multi-omics integration in the discovery cohort

A multicomponent omics model was constructed on a subset of 17 patients from the discovery cohort who had all omics data available. The early integrated dataset included 1136 features: 138 serum metabolites, 433 serum lipids, 172 stool metabolites, 185 stool lipids, 201 microbi

al taxonomies and 7 disease activity and demographic characteristics. Following the feature selection within each omic layer, a total of 30 features were retained to the optimization step. These included 5 clinical features, 5 serum metabolomics, 5 serum lipidomics, 5 fecal metabolomics, 5 fecal lipidomics, and 5 microbial components. The final multi-omics model included 9 features from all omics layers, excluding clinical features, which did not contribute to the model. Specifically, higher levels of serum metabolite 2-hydroxyglutaric acid, bacterial family *Bifidobacteriaceae,* fecal 3-methyladipic acid, and serum lipid dodecanedioylcarnitine (C12DC) were observed in responders. In contrast, nonresponders had higher levels of serum lipids Cer (d18:0/22:0) and HexCer(d18:1/d26:1), fecal lipid DG(16:0 20:0), fecal lipid PC aa 42:2, and the microbiome genus *CAG-56* (Figure 3A). The final integrated model demonstrated training performance of 94% accuracy (95% CI, 82%-100%; Table S1). Consistently, the PCA plot showed a clear separation between EEN responders and nonresponders (Figure 3B).

## Validation cohort

Finally, we evaluated the performance of both the integrated multi-omics and the single-omics models using an independent external validation cohort. The multi-omics model was successfully replicated, with an AUROC of 0.81 (95% CI, 0.60%-1.00), 81% accuracy (95% CI, 62%-95%), 56% sensitivity (95% CI, 22%-88%), and 100% specificity (95% CI, 100%-100%). The PPV was 100% and the NPV was 75% (95% CI, 53%-94%), indicating that all patients predicted to be nonresponders to nutritional therapy did not respond, while 75% of those predicted to be responders indeed responded (Table 2). The PCA analysis further supported the multi-omics model accuracy with a clear separation between responders and nonresponders (Figure 3B). Most single-omics models, including those based on clinical variables, serum and fecal metabolomics and microbiome data, did not achieve sufficient predictive performance in the external validation cohort (AUROC range, 0.24-0.56). In contrast, serum and fecal lipidomics models demonstrated fair predictive utility, with AUROCs of 0.74 and 0.71 (95% CI, 0.47-0.95 and 0.46-0.93,

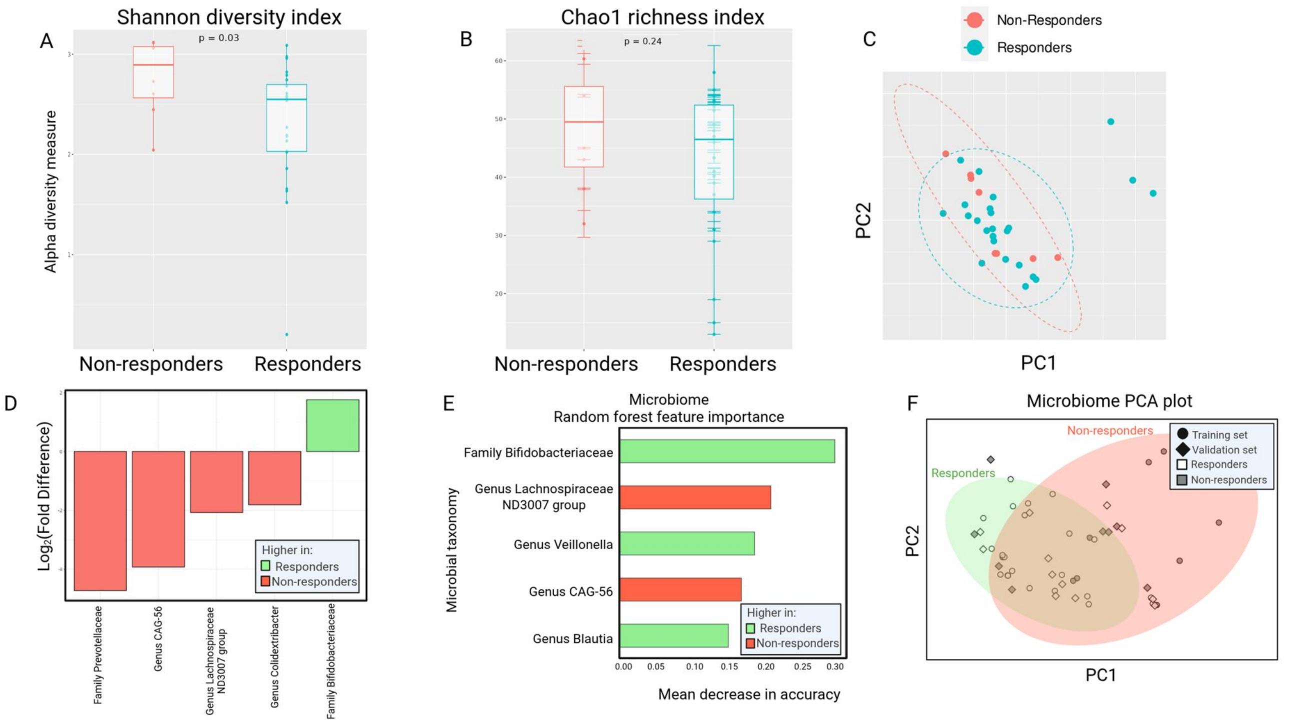


**Figure 2** Microbiome diversity, differential abundance and predictive features associated with EEN response in CD. (A-B) Microbiome alpha-diversity, (A) Shannon diversity index and (B) chao1 richness index shows bacterial taxonomic genus diversity and richness for EEN response; (C) Microbiome beta-diversity calculated using Bray-Curtis distance using Principal Coordinate Analysis plot; (D) Differential abundance of bacterial taxonomies between EEN responders and nonresponders using MaAsLin2; (E) Feature importance calculated via machine learning random forest model of the features retained by the mRMR algorithm; In the bar plots, green means higher median values in responders, and red higher median values in nonresponders. (F) PCA plot displaying PCA results for the microbiome results. In the PCA plots, each point represents a subject positioned in a 2-dimensional space defined by the first 2 principal components (PC1 and PC2); responders in white and nonresponders in gray, circles are discovery cohort subjects and diamonds are validation cohort subjects. Ellipses are centered at the group means, with dimensions based on twice the standard deviation of data within each group (responders in green and nonresponders in red). EEN, exclusive enteral nutrition; MaAsLin2, Microbiome Multivariable Associations with Linear Models; mRMR, minimum redundancy maximum relevance; PCA, principal component analysis.

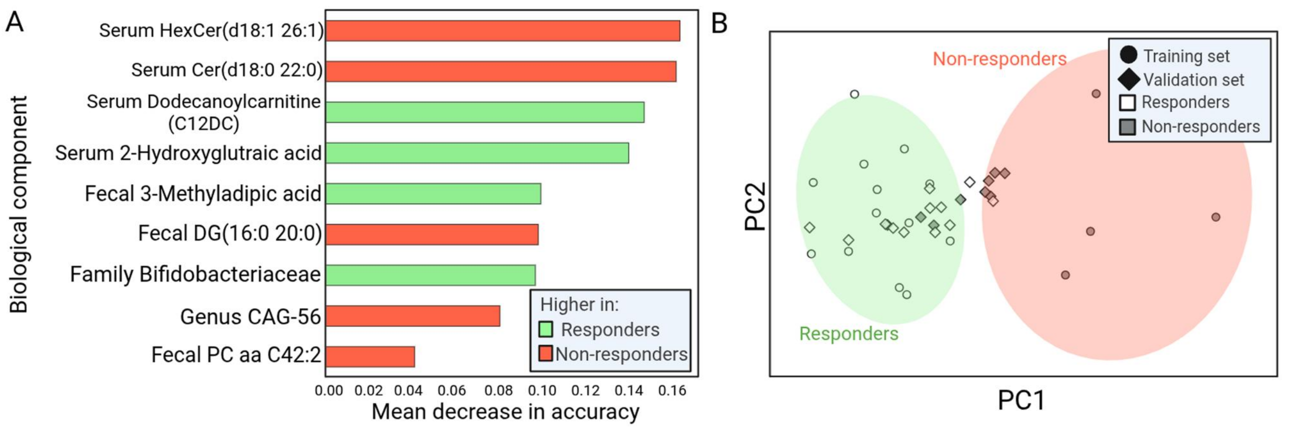


**Figure 3** Multi-omics random forest model with retained feature from all single-omics to predict EEN response in CD. The multi-omics model was based on serum metabolomics, serum lipidomics, fecal metabolomics, fecal lipidomics and fecal microbiome composition. (A) Feature importance calculated via machine learning random forest model of the features retained by the mRMR algorithm; In the bar plots, green means higher median values in responders, and red, higher median values in nonresponders. (B) PCA plot displaying PCA results for the multi-omics results. In the PCA plots, each point represents a subject positioned in a 2-dimensional space defined by the first 2 principal components (PC1 and PC2); responders in white and nonresponders in gray, circles are discovery cohort subject and diamonds are validation cohort subjects. Ellipses are centered at the group means, with dimensions based on twice the standard deviation of data within each group (responders in green and nonresponders in red). CD, Crohn disease; EEN, exclusive enteral nutrition; mRMR, minimum redundancy maximum relevance; PCA, principal component analysis

respectively; Table S1). When directly comparing lipidomics models to the integrated multi-omics model, serum lipidomics alone showed fair training performance in the discovery cohort and robust predictive ability in the validation cohort (serum lipidomics: AUROC 0.77 and 0.74 [95% CI, 0.58-0.94 and 0.47-0.95], accuracy 74% and 76% [95% CI, 59-88% and 57-95%]; multi-omics: AUROC 1.00 and 0.81 [95% CI, 1.00-1.00 and 0.60-1.00], accuracy 94% and 81% [95% CI, 82-100% and 62-95%]; training and validation, respectively; Table S1 and Table 2). Fecal lipidomics also demonstrated consistent, albeit lower, performance in both cohorts (AUROC 0.67 and 0.71 [95% CI, 0.27-0.98 and 0.46-0.93], accuracy 74% and 67% [95% CI, 53%-89% and 43%-86%], respectively; Table S1 and Table 2).

## Discussion

In this study, we demonstrate the proof-of-concept for using an ML model on multi-omics data to predict therapeutic response to EEN in pediatric CD. An integrated multi-omic ML model combining serum and fecal metabolomics, serum and fecal lipidomics, and microbiome composition achieved superior and more robust performance than each of the single omic models separately, with an AUROC of 0.81 (95% CI, 0.6-1.0) in the external validation cohort, showing a realistic estimate for the model's generalizability and robustness in concordance with the training performance. These findings highlight the potential of multi-omics integration to enhance individualized treatment planning in IBD. Among the single-omic models, serum metabolomics, fecal metabolomics, gut microbiome, and serum lipids showed predictive utility in the discovery cohort (AUROC values, 0.76-0.90), but only serum lipidomics was replicated in the external validation cohort with fair to moderate performance.

The discovery cohort was assembled over several years with identical eligibility criteria, sample collection procedures, and EEN management throughout this period, ensuring methodological consistency across the study. The observed 66% response rate in the discovery cohort and 57% in the validation cohort reflects real-life clinical practice, in which variability in adherence commonly results in somewhat lower effectiveness compared with tightly controlled clinical trials. Nevertheless, this rate remains within the range reported in established pediatric literature, in which approximately 60%-80% of children achieve remission with EEN.

Lending support to the biological plausibility of our findings, the serum (host) lipidomics model retained features associated with known inflammatory mechanisms. Nonresponders had elevated levels of ceramides (eg, HexCer[d18:1/26:1], Cer[d18:0/24:0], and Cer[d18:0/22:0]), a lipid class implicated in the regulation of immune responses, intestinal barrier integrity, and metabolic stress.[23] Ceramides have been shown to promote inflammatory signaling cascades and disrupt epithelial barrier function, contributing to mucosal inflammation in IBD.[24] These findings suggest that the baseline serum lipid profile may reflect systemic inflammation and metabolic activity, which could help explain differences in response to nutritional therapy.

Additionally, several serum and fecal metabolites that contributed to the multi-omics model reflect host and microbial metabolic states. Among these, serum 2-hydroxyglutaric acid, elevated in responders, has been associated with mitochondrial function and gut microbial metabolism, potentially indicating a more favorable metabolic environment.[25–27] Another retained feature, fecal 3-methyladipic acid, is a dicarboxylic acid linked to fatty acid oxidation and has been reported as a microbial co-metabolite.[28,29] Together, these metabolites underscore the contribution of energy metabolism and host-microbial cross-talk. When integrated with additional omic layers (eg, lipidomic and microbiome signals), they illustrate how multi-omics models can capture subtle, yet complementary, molecular signatures associated with biological mechanisms and therapeutic outcomes. The microbiome component, though individually less predictive, provided complementary information within the integrated model. Enrichment of *Bifidobacteriaceae* in responders is consistent with its known role in short-chain fatty acid production, improved mucosal integrity, and anti-inflammatory activity.[30,31] In contrast, genera such as the *Lachnospiraceae ND3007 group* and *CAG-56* (within the *Lachnospiraceae* family), as well as taxa from the *Prevotellaceae* family, which are linked to dysbiosis and inflammatory states, were more abundant in nonresponders.[2,32,33] Collectively, these host and microbial signatures converge on pathways related to epithelial barrier maintenance, lipid metabolism, and immune modulation—highlighting the mechanistic coherence and translational relevance of the identified predictive features.

Interestingly, in our cohort, children who did not respond to EEN had higher α diversity (Shannon index) at baseline. This finding is somewhat unexpected, as greater microbial diversity is often associated with gut health. Several studies have reported that EEN reduces microbial diversity during treatment, particularly among patients who achieve remission,[34–36] suggesting that a simpler, diet-sensitive microbiome may be more amenable to modulation. However, baseline diversity trends have varied across studies: Schwerd et al. observed no significant association between diversity and EEN response,[37] while other investigators have reported lower baseline diversity in nonresponders.[12] These inconsistencies highlight the complexity of host-microbiota interactions in IBD and the need for further research to clarify the role of microbial diversity in predicting treatment outcomes.

By capturing complementary signals from host and microbial metabolism, lipid profiles, and gut microbiota, the multi-omics model yielded promising predictive power in the validation cohort, supporting its potential for developing real-world clinical biomarkers. Among the individual omics layers, serum lipidomics alone demonstrated robust predictive ability in the validation cohort. The performance was slightly lower than that achieved by the multi-omics model but remained clinically meaningful. This finding suggests that serum-based assays may provide a practical lower cost alternative in settings in which full multi-omics profiling is not feasible. In contrast, fecal lipidomics showed a weaker and less consistent signal and therefore appears insufficient as a standalone clinical tool. Ultimately, serum-based lipidomics testing is minimally invasive, technically mature, and cheaper and more easily standardized than multilayer omics panels, which currently require specialized assays and longer turnaround times.

Lipidomics was the strongest-performing modality within each sample type. While serum lipid profiles reflect systemic inflammatory and metabolic activity and yielded a fair and stable signal in the reported models, fecal lipidomics, shaped also by microbial metabolism, luminal dynamics, and sampling variability, exhibited a limited signal. This divergence is consistent with findings from other multi-omics studies in IBD, in which serum sphingolipid and ceramide pathways consistently emerge as robust inflammatory markers,[23,24] and lipidomics contributes to predictive biomarker identification.[38] Taken together, these observations support the role of lipidomics, particularly from serum, as a biologically coherent, technically feasible single-omics layer to predict response to nutritional therapy in CD and an important contributor within multi-omics prediction frameworks.

As the predictive modeling in this study is exploratory, further work is required before clinical implementation can be considered. Key next steps are validation in independent larger cohorts and the standardization of multi-omics measurements. This further step may include targeted assays for selected metabolites and harmonized pipelines for microbiome profiling, to enable development of a reproducible multi-omics panel.

Our findings build upon previous work in this area. Nichols et al.[12] demonstrated that gut metabolome and microbiota signatures could be used to distinguish responders from nonresponders to EEN in pediatric CD. Jones et al.[13] demonstrated a strong predictive utility of microbial species alongside clinical features in predicting response to EEN in 22 children with CD. Both studies focused primarily on microbial composition combined with either fecal metabolites or clinical features and similarly showed enrichment of short-chain fatty acid–associated taxa (*Ruminococcaceae*). In our study, fecal metabolites and a different taxon associated with short-chain fatty acid metabolism (*Bifidobacteriaceae*) also contributed to this classification, but serum sample features, such as serum metabolites (eg, 2-hydroxyglutaric acid) and lipids (eg, ceramides) provided complementary information, improving model robustness and generalizability when integrated in a multi-omics framework, combining both host and microbial signals.

This proof-of-concept study has several limitations. First, both cohorts were small, which restricted the statistical power of the analyses, which are therefore considered exploratory. Certain variables, such as fecal calprotectin at follow-up, were not available for all patients. Furthermore, data completeness across omic layers was partial. To that end, the use of cross-validation for feature selection and replication of the findings in the external validation cohort mitigated the risk of type 1 error. Second, the heterogeneity of multi-omics data and the need for exclusion of metabolites detected in fewer than 25% of samples for reducing noise may have resulted in low-abundance relevant features being omitted. Moreover, dimensionality reduction in a small sample further increases the risk of overfitting, even when feature selection was performed within cross-validation folds. The observed accuracy and effect sizes should therefore be interpreted with caution and confirmed in larger, prospective, multicenter cohorts. Despite these constraints, the consistency of predictive patterns across independent cohorts, recognizing that no predictive model should be adopted without external validation[39], supports the feasibility of multi-omics integration as a promising step toward precision nutrition in pediatric Crohn's disease.

Overall, our findings demonstrate the potential of combining multi-omics data with ML to construct clinically relevant predictive models, even in modestly sized real-world cohorts. By integratin signals from host and microbial metabolism, lipid profiles, and the gut microbiome, such models can help guide early therapeutic interventions, such as identifying patients more or less likely to respond to treatment. We provide a proof-of-concept scheme for using multi-omics analyses in personalized nutritional treatment in IBD, but future explorations are required in larger cohorts to further verify our findings.

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# Author contributions

A.A.: Conceptualization, data curation, formal analysis, investigation, methodology, software, validation, visualization, writing–original draft, and writing–review & editing. L.G.: Writing–original draft, and writing–review & editing. Y.A.-F.: Conceptualization, data curation, writing–original draft, and writing–review & editing. G.F.: Conceptualization, methodology, project administration, supervision, funding acquisition, and writing–review & editing. Y.T.: Consultation, methodology, and writing–review & editing. E.B.: Supervision, methodology, and writing–review & editing. L.P.: Conceptualization, data curation, writing–original draft, and writing–review & editing. E.O.-M.: Data curation, medical follow-up, methodology, and writing–review & editing. R.L.-T.: Data curation, medical follow-up, methodology, and writing–review & editing. O.L.: Data curation, medical follow-up, methodology, and writing–review & editing. D.Y.: Data curation, medical follow-up, methodology, and writing–review & editing. A.A.: Data curation, medical follow-up, methodology, and writing–review & editing. Efrat Broide: Data curation, validation and writing–review & editing. A.Y.-F.: Data curation, validation, and writing–review & editing. J.K.: Data curation, validation, and writing–review & editing. T. S.: Data curation, validation, and writing–review & editing. E.W.: Data curation and writing–review & editing. D.T.: Conceptualization, data curation, funding acquisition, investigation, methodology, supervision, writing–original draft, and writing–review & editing.

# Supplementary material

Supplementary data are available at *Inflammatory Bowel Diseases* online.

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# Conflicts of interest

A.A.—no conflict of interest to declare. L.G.—no conflict of interest to declare. Y.A.F.—no conflict of interest to declare. G.F.—received in the past 3 years consultation fee from Lilly outside the submitted work. Y.T.—no conflict of interest to declare. E.B.—no conflict of interest to declare. L.P.—no conflict of interest to declare. E.O.M.—no conflict of interest to declare. R.L.T.—no conflict of interest to declare. O.L.—no conflict of interest to declare. D.Y.—no conflict of interest to declare. A.A.—no conflict of interest to declare. E.B.—no conflict of interest to declare. A.Y.F.—no conflict of interest to declare. J.K.—received grant from Nestle outside the submitted work. T.S.—received lecture honoraria from Nutricia, MSD and Abbvie; Travel support from Abbvie and Ferring; Consulting fees from AstraZeneca. All outside the submitted work. E.W.—received personal fees from Janssen, AbbVie, Nestle Health Sciences, Mead Johnson Nutrition, Pfizer, BioJamp outside the submitted work. D.T.—received consultation fees from Janssen, Pfizer, Ferring, Abbvie, Takeda, Prometheus Biosciences, Lilly, SorrisoPharma, Boehringer Ingelheim, Galapagos, BMS, AlfaSigma, Merck, Gentech; Research support from Janssen, Abbvie, Takeda, Pfizer; Royalties from Shaare Zedek Medical Center, Hospital for Sick Children, all outside the submitted work.

# Data availability

The raw multi-omics datasets generated for the current study have been deposited in Zenodo with code and are publicly available [DOI 10.5281/zenodo.17541258].

# Research ethics

This study and the validation study were approved by The Shaare Zedek Helsinki Committee and all participants provided written informed consent upon enrollment. Discovery cohort (Biobank) approval number 74-13 and validation cohort (Tasty&Healthy) approval number 294-19.

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